Desirability Validation Report: Precision Fertilizer Optimization Platform

Executive Summary

Market Opportunity: Corn Belt farmers face $100K+ annual fertilizer spend with nitrogen prices 6-20% higher in 2025, creating acute margin pressure on thin-margin operations.

Problem Validation: Five expert advisors confirmed fertilizer cost volatility and suspected over-application are real pain points, but desirability remains unvalidated—farmers have not been interviewed.

Customer Desirability: Core question: Will farmers actually *pay* for a solution before seeing peer proof? This is unconfirmed and critical to validate before product investment.

Overall Assessment: Strong market timing and clear problem definition. However, **desirability is unproven**. Recommend conducting 15-20 farmer problem interviews immediately to validate willingness-to-pay and behavior change openness. Confidence Level: MODERATE on problem, LOW on desirability.

Validation Criteria

Evaluation Framework: Three-round expert debate structured around core desirability question: Will farmers experiencing fertilizer cost/optimization problems actually pay for a solution?

Six Core Dimensions to Validate: - Problem Intensity: Acute pain from cost volatility and suspected over-application - Current Solution Satisfaction: Frustration with agronomist advice, flat-rate recommendations, or spreadsheets - Behavior Change Openness: Willingness to apply different rates by field - Proof Requirements: What evidence threshold triggers payment commitment - Agronomist Influence: Is endorsement a blocker or accelerator - Willingness to Pay: Actual commitment to budget in price-spike years

Data Sources: Expert panel debate (Maris-Customer Insights, Baylen-Growth, Isla-Industry Trends, Delmar-Behavioral Economics, Delta-Investor). Business Canvas, Competitor Landscape, Customer Behavior Story.

Areas of Expert Agreement

- **Problem is Real:** All experts confirmed fertilizer cost volatility and suspected over-application resonate as genuine pain points in Corn Belt.
- **Farmer Behavior is Risk-Averse:** Consensus that yield protection is ranked higher than cost savings; loss aversion dominates farmer decision-making.
- **Agronomist is Accelerator, Not Blocker:** All agreed agronomist endorsement accelerates adoption but isn't a must-have for early adopters.
- **Desirability Unproven:** Complete agreement that market research and competitive analysis cannot substitute for direct farmer conversation on willingness-to-pay.
- **Early Adopter Segment Exists:** Experts agreed a subset of 500-2,000 acre pragmatists will take calculated risks; not all farmers in segment are equally motivated.

Areas of Expert Disagreement

- **Interview Sequencing:** Maris advocated problem-first (open-ended), then solution validation. Delmar recommended showing example prescriptions early to test *revealed* preferences vs. stated preferences.
- **Segment Definition:** Delta challenged pre-defined "pragmatic margin protector" segment, arguing interviews should be broad first to discover *actual* high-motivation segment. Founder agreed—interviews will be broad.
- **Willingness-to-Pay Timing:** Delmar emphasized anchoring price questions to 2025 price-spike context for realism. Baylen wanted budget allocation questions earlier to assess ag tech spending patterns.
- **Geographic Focus:** Isla recommended spanning 2-3 geographies and crops (corn vs. wheat) to validate demand variation. Founder's focus was Iowa/Illinois corn initially; Isla's input suggests broader pilot.

Unresolved Critical Questions

- What % of farmers in target segment experience acute fertilizer cost pain (vs. accepting it as status quo)?
- What % would actually *apply* field-specific rates recommended by software, or just intellectually agree it's good idea?
- At what proof threshold do farmers commit to paying ($150/month) without peer validation—university research alone, or peer farm results required?
- Is demand durable (structural soil variability and optimization need) or cyclical (driven only by temporary 2025 price spike)?
- Do agronomists actively hear demand from farmers for optimization tools, or is demand primarily farmer-initiated?
- Are most motivated farmers within your defined 500-2,000 acre segment, or do high-motivation outliers exist in smaller/larger operations?

Recommended Interview Approach

Target: 15-20 farmers across Iowa, Illinois, Indiana (corn and wheat if possible).

Sampling: Deliberately broad first (don't pre-filter by farm size or tech adoption). Cast wide net to discover who truly has acute problem vs. mild frustration.

Interview Structure: 1. Open-ended problem exploration (current fertilizer decision process, pain points, costs) 2. Behavior change openness (have you tried optimizing rates? what would trigger action?) 3. Proof requirements (what evidence makes you trust a new tool?) 4. Willingness-to-pay (in 2025 price context, would you commit budget?) 5. Agronomist role (how important in your decisions?)

Success Metric: 30%+ of farmers show strong signals on all five dimensions = desirability validated. Below 30% = reassess positioning or market focus.

High-Risk Assumptions

- **Assumption:** Farmers' suspected over-application translates to willingness to change. **Risk:** Farmers may intellectually agree but resist actual behavior change due to yield fear. **Mitigation:** Test with example prescriptions; measure revealed preferences, not stated preferences.
- **Assumption:** $10-30/acre savings justifies $150/month subscription. **Risk:** Farmers may perceive savings as insufficient or untrusted without peer proof. **Mitigation:** Validate willingness-to-pay in interviews; test loss-framing vs. gain-framing.
- **Assumption:** Demand persists beyond 2025 price spike. **Risk:** Cyclical demand collapses when prices normalize. **Mitigation:** Separately validate structural (soil variability) vs. cyclical (price) drivers in interviews.
- **Assumption:** Early adopters reachable via agronomist partnerships. **Risk:** Agronomists lack incentive to promote margin-transparent tools. **Mitigation:** Identify which farmers are willing to adopt without agronomist—they're your Year 1 pilots.

Success Metrics & Validation Targets

Desirability Validation Thresholds (Post-Interview): - Problem Intensity: 70%+ report fertilizer cost volatility as significant business concern - Behavior Change Openness: 50%+ have experimented with variable rates or express clear willingness to try - Willingness to Pay: 30%+ commit to paying $100-200/month in price-spike scenarios without peer proof required - Proof Sufficiency: Identify proof threshold (peer farm data, university research, or trial on single field) that 30%+ deem acceptable - Agronomist Influence: Document % for whom agronomist endorsement is must-have vs. nice-to-have - Demand Durability: Separately validate structural (soil optimization) vs. cyclical (2025 prices) drivers

Go/No-Go Decision Point: If fewer than 30% show strong desirability signals across all dimensions, pause product development and reassess market positioning, segment focus, or value proposition before scaling.